

MEASUREMENT

— Length

Session 1 · 30 Minutes · Beginner

Category	Subject	Session	Duration
Measurement – Length	Mathematics	Session 1 of 3	30 Minutes

Board	Depth	Language
CBSE / NCERT (ICSE compatible)	Beginner	English (with Hindi transliterations)

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SESSION HOOK & WARM-UP (0:00 – 0:03)

Opening Story: Riya's Dupatta Question

Begin the session with this relatable Indian scenario to instantly capture attention and activate prior knowledge about measurement in daily life.

The Story	Riya wants to buy a new dupatta for her doll. The shopkeeper asks — How long should it be? How does Riya tell him?
Discussion	Ask the class: Can you think of things we measure every day? (Cricket bat, braid, the doorway, a water bottle, a chapati rolling pin...)

LEARNING OBJECTIVES

By the end of this session, every learner will be able to:

LO 1 — Conceptual	Understand that length tells us how long, tall, or wide an object is, from one end to the other.
LO 2 — Non-Standard	Measure everyday objects using non-standard units such as hand spans, finger widths, pencil lengths, and footsteps.
LO 3 — Standard Unit	Recognise the centimetre (cm) as the standard unit of length and use a ruler correctly to measure objects up to 30 cm.
LO 4 — Comparison	Compare the lengths of two or more objects using: longer, shorter, taller, wider, equal / same length.
LO 5 — Estimation	Make a reasonable estimate of an object's length before measuring it.
LO 6 — Application	Apply length measurement to solve simple, relatable problems from everyday Indian contexts.

Bloom's Taxonomy Mapping

Bloom's Level	Learning Objective
Remember	LO 3 — Recall cm as standard unit
Understand	LO 1 — Grasp concept of length
Apply	LO 2, LO 3 — Use ruler / hand span
Analyse	LO 4 — Compare lengths

Evaluate	LO 5 — Estimate before measuring
Create	LO 6 — Solve real-world problems

30-MINUTE SESSION OUTLINE

Time	Segment	Activity / Content
0:00 – 0:03	Hook / Warm-Up	Riya's Dupatta scenario · Ask: "What things do we measure every day?" · Show cricket bat, palm tree, thali images
0:03 – 0:08	Concept Introduction What is Length?	Show images: cricket bat (LONG), palm tree (TALL), thali (WIDE) · Introduce vocabulary: long, short, tall, wide, same length
0:08 – 0:15	Non-Standard Measurement (7 min)	Measure desk with hand span · Measure pencil box with a pencil · Count footsteps door → desk · Arjun vs Priya hand-span problem — bridge to cm
0:15 – 0:22	Standard Unit — cm (7 min)	Introduce cm with ruler · 4-step ruler guide · Show: Parle-G biscuit $\approx$ 5 cm, Apsara pencil $\approx$ 17 cm, 10-rupee note $\approx$ 12 cm · Common mistakes (start from 1, slanted ruler)
0:22 – 0:27	Compare + Estimate (5 min)	Show pencil vs crayon side by side · Estimation challenge: "Guess the eraser length!" · Reveal actual length
0:27 – 0:30	Recap + Quick Check (3 min)	3-question quiz (unit? / start at? / longer: 20 cm or 15 cm?) · Summary chant: "Measure it right, measure it well — in centimetres, we always tell!"

KEY CONCEPTS WITH INDIAN EXAMPLES

Concept 1 — What is Length?

Definition: Length tells us how long, tall, or wide something is, from one end to the other.

LONG	A cricket bat is LONG from handle to blade.
TALL	A palm tree is TALL — it reaches up high!
WIDE	A thali (steel plate) is WIDE — it stretches across.

Comparing Words	Form	Equal
long / short	long → longer → longest short → shorter → shortest	same length = equal Two things can be the same size!
tall / wide	tall → taller → tallest wide → wider → widest	

Concept 2 — Non-Standard Measurement

Definition: Measuring using everyday objects that are NOT fixed in size — like a hand span, finger, pencil, or paper clip. Builds intuition before standard units.

Example 01	My notebook is 3 hand spans long.
Example 02	The diya (lamp) on the puja table is 2 finger widths tall.
Example 03	The steel tiffin box lid is about 4 pencil widths across.

■ *Bridge to Standard: Arjun's hand span is bigger than Priya's. Same table — different answers! This shows WHY we need a unit everyone agrees on — the centimetre!*

Concept 3 — Standard Unit: The Centimetre (cm)

Definition: A centimetre (cm) is a standard unit of length. It is the same everywhere in the world. Abbreviation: **cm**.
Tool used: Ruler (scale).

Indian Object	Approximate Length
Apsara pencil (new)	≈ 17 cm
HB pencil	≈ 19 cm
10-rupee note	≈ 12 cm
Parle-G biscuit width	≈ 5 cm
Classroom ruler	15 cm or 30 cm

Notebook page width	≈ 17 cm
Width of one finger	≈ 1 cm

HOW TO USE A RULER CORRECTLY

Step 1	Place the ruler flat on the object.
Step 2	Line up one end of the object with the 0 mark — NOT the edge of the ruler.
Step 3	Look straight down (not from the side) at where the other end of the object reaches.
Step 4	Read the number — that is the length in cm .

COMMON MISTAKES TO AVOID

X Wrong Way	✓ Right Way
Starting from 1 instead of 0 on the ruler	Always start from the 0 mark
Holding the ruler slanted or not straight	Keep the ruler flat and straight along the object
Reading from the wrong side of the ruler	Read from the correct side — where numbers are visible

COMPARING LENGTHS

Example 01	A bindi is shorter than a pencil.
Example 02	A sugarcane is longer than a ruler.
Example 03	Raju is taller than Meena.
Example 04	A samosa is wider than a Parle-G biscuit.
Example 05	Two identical colour pencils have the same / equal length .

ESTIMATION

Definition: Guessing the length of an object **BEFORE** measuring it. Estimation develops number sense and real-world maths thinking.

Object (Estimation Anchor)	Approx. Length
Width of one finger	≈ 1 cm
Width of thumb	≈ 2 cm
Parle-G biscuit width	≈ 5 cm
Small/used pencil	≈ 10 cm
New Apsara pencil	≈ 17 cm
30 cm ruler (geometry box)	30 cm

INDIAN CONTEXT OBJECT BANK

Category	Objects for Activities
Classroom	Chalk piece, duster, blackboard ruler, notebook, pencil, eraser, geometry box, colour pencil, sharpener
Home/Kitchen	Steel tiffin (dabba), chapati rolling pin (belan), ladle (karchi), spoon, broom (jhadoo), water bottle, soap bar
Food	Parle-G biscuit, banana, carrot, drumstick (murungakkai), sugarcane piece, roti, idli
Cultural	Diya (lamp), bindi, rangoli design width, saree border width, agarbatti (incense stick)
Sports	Cricket bat, carrom striker, kabaddi court line, skipping rope
Nature	Neem leaf, mango leaf, flower petal, earthworm (for comparison)

QUICK QUIZ — CHECK YOUR UNDERSTANDING

Q	Question	Answer
1	What unit do we use to measure length?	Centimetre (cm)
2	Where does ruler measurement start?	From 0 (zero mark)
3	Which is longer: 20 cm or 15 cm?	20 cm

SUCCESS CRITERIA

A learner has successfully completed this session if they can:

- ■ Explain what "length" means in their own words.
- ■ Measure an object using hand spans or finger widths and report the number.
- ■ Correctly place a ruler at 0 and read a measurement in whole centimetres.
- ■ Identify which of two objects is longer / shorter when shown side by side.
- ■ Make a reasonable estimate (within ± 5 cm) for a common object.
- ■ Solve one simple word problem involving length with an Indian context.

Threshold	Result
4 out of 6 criteria met	Session Objective Achieved ✓
6 out of 6 criteria met	Mastery ■

VOCABULARY GLOSSARY

Term	Simple Definition
Length	How long something is from one end to the other.
Measure	To find out how long, tall, or wide something is.
Ruler	A flat tool used to measure length, marked in cm.
Centimetre (cm)	A small standard unit of length. 100 cm = 1 metre.
Non-standard unit	Using everyday things (hand, pencil) to measure — not a fixed unit.
Estimate	A careful guess before measuring.
Longer / Shorter	Has more / less length than another object.

Taller / Wider	Has more height / width than another object.
Same length	Two things that measure exactly the same.
Zero mark	The starting point (0) on a ruler.

RECAP & SUMMARY

What We Learned Today!

Point 1	Standard unit = centimetre (cm) — same everywhere in the world!
Point 2	Always start measuring from 0 on the ruler — never from the edge or from 1!
Point 3	Length tells us how long, tall, or wide something is.
Point 4	Non-standard units (hand span, finger, pencil) help build intuition for length.
Point 5	Estimation = a careful guess before measuring — within ± 5 cm is great!

"Measure it right, measure it well — in centimetres, we always tell!"

NEXT SESSIONS PREVIEW

Session	Title	Key Topics
Session 2 (Intermediate)	Measuring in Metres	Introduce metre (m) · $100\text{ cm} = 1\text{ m}$ · Measure classroom wall, corridor, door height · Saree length, cricket pitch
Session 3 (Intermediate)	Word Problems & Real-Life Applications	Multi-step problems · Order objects shortest → longest · Draw lines of given lengths · Home project: "Measure 5 things"